Walter Ac-Pangan

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EMPLOYMENT

Department of Agricultural and Applied Economics, Virginia Tech
Agribusiness Instructor 2025 – Present

Department of Agricultural Economics, Kansas State University
Graduate Research Assistant 2019 – 2025

Department of Hort Science, University of Florida
Research Scholar 2018

Department of Applied Economics, University of Minnesota
Visiting Scholar 2017

EDUCATION

Ph.D. Agricultural Economics 2025
Kansas State University (Manhattan, KS)
Dissertation: Essays on Forecasting and International Trade in Grain Markets
Committee Chair: Nathan P. Hendricks

M.Sc. Agricultural Economics 2021
Kansas State University (Manhattan, KS)
Thesis: How do changes in market fundamentals affect the hedging in US live cattle markets?

B.Sc. Agribusiness Management 2017
Zamorano University, Pan-American Agricultural School (Honduras)

RESEARCH WORKS IN PROGRESS

• **Ac-Pangan, W.**, Ortez, M., Zhang, Y., Cary, M. In-Season Forecasting of Corn Yields Using Remote Sensing Data. (In Progress).

• **Ac-Pangan, W.**, Peña-Levano, L., French, E. Identifying Sources of Productivity Differences in Automated Milking Systems. (In Progress).

• **Ac-Pangan, W.** Cocoa Commodity Market Analysis: Supply, Prominent marketing, and Price Discovery Mechanisms (In progress).

• **Ac-Pangan, W.** Coffey, B.K. An Analysis of Live Cattle Basis Prediction Errors. (In Progress)

SUBMITTED MANUSCRIPTS FOR PUBLICATION

• **Ac-Pangan, W.**, Hendricks, NP. Effects of Non-Tariff Trade Barriers in Grain Markets: The Case of Rice Export Ban Imposed by India. (*Submitted to Applied Economic Perspective and Policy*).

• Jo, Francis. Reyes, P. Hendricks, NP., **Ac-Pangan, W.** A Unified Algorithm For Penalized Likelihoods In Model Selection Of Spatial-Temporal Models For Lattice Data. (*Submitted to Journals of the Royal Statistical Society*).

• **Ac-Pangan, W.**, Hendricks, NP. In-Season US Crop Acreage Forecasting Using Machine Learning. (*Submitted to International Journal of Forecasting*).

• Sharma, P., Kim, J.,**Ac-Pangan, W.,** Peterson, H., Shanoyan, A., Yu, J. Identifying Gaps in Food Manufacturing Business: Local Drivers and Impediments for Small Agri-Food Businesses. (*Revised and Resubmitted to Applied Economic Perspective and Policy*).

• **Ac-Pangan, W.** Hendricks, N., Zereyesus, Y. A., Kee, J. Forecasting Grain Supply of Low and Middle-Income Countries. (*Revised and Resubmitted to Agricultural and Resource Economics Review*).

PEER-REVIEWED PUBLICATIONS

• Rossi, L., Hallman, L. M., Adams, S. N. **Ac-Pangan, W.** (2020). Impact of a Soil Conditioner Integrated into Fertilization Scheme on Orange and Lemon Seedling Physiological Performances. Plants, 9(7), 812. Available at: https://doi.org/10.3390/plants9070812

• Adams, S. N., **Ac-Pangan, W.**, Rossi, L. (2019). Effects of Soil Salinity on Citrus Rootstock 'US- 942' Physiology and Anatomy, HortScience horts, 54(5), 787-792. Available at: https://doi.org/10.21273/HORTSCI13868-19

OTHER PUBLICATIONS

• Zereyesus, Y. A., Kee, J., Hendricks, NP., **Ac-Pangan, W.**, Cardell, L., Nava, N.J., Jellife, J., Morgan, S. (2023). Special Article: Projecting Long-Term Staple Grain Production for the International Food Security Assessment: A Time Series Cross-Validation Approach. USDA ERS. International Food Security Assessment, 2023-2033. Page 38-47. Available at: https://www.ers.usda.gov/webdocs/outlooks/107271/gfa-34.pdf?v=916.1

• **Ac-Pangan, W.** Plan de negocios para el establecimiento de una exportadora de cacao fino de aroma en Cobán, Alta Verapaz, Guatemala. Zamorano University Wilson Popenoe Library. Available at: https://bdigital.zamorano.edu/bitstream/11036/6136/1/AGN-2017-001.pdf

ACADEMIC PRESENTATIONS

• **Ac-Pangan, W.**, Hendricks, NP. (2026). Machine Learning Approaches for In-Season Forecasting of Corn Yield and Acreage. Selected paper will be presented at the Australasian Agricultural and Resource Economics Society (AARES). Feb 8, 2026. Adelaide, Australia.

• **Ac-Pangan, W.**, Hendricks, NP. (2025). Effects of Non-Tariff Trade Barriers in Grain Markets: The Case of Rice Export Ban Imposed by India. Selected paper presented at the Australasian Agricultural and Resource Economics Society (AARES). Feb 14, 2025. Brisbane, Queensland, Australia.

• **Ac-Pangan, W.**, Sharma, P., Kim, J. (2024). Drivers of the Establishment of New Local Food Manufacturing Businesses in the Heartland Region. USDA-AMS Local Food Economics Data Visualization Challenge presented at Agricultural and Applied Economics Association Meeting (AAEA), July 28, 2024. New Orleans, Louisiana.

• **Ac-Pangan, W.**, Hendricks, N., Zereyesus, Y. A., Kee, J., Morgan, S., Jellife, J., Nava, N., Cardell, L. (2023). Global grain supply forecasting. Selected paper presented at Agricultural and Applied

Economics Association Meeting (AAEA), July 24, 2023. Washington, DC. https://ageconsearch.umn.edu/record/335783?ln=en

• Zereyesus, Y. A., Kee, J., Morgan, S., Jellife, J., Nava, N., Cardell, L., Hendricks, N., **Ac-Pangan, W.** (2023). Does trade openness affect global grain supply response? Selected paper presented at Agricultural and Applied Economics Association Meeting (AAEA), July 24, 2023. Washington, DC. https://ageconsearch.umn.edu/record/335666?ln=en

• **Ac-Pangan, W.**, Hobbs, L., Sharma, P., Ofori-Bah, C., Sharma, M., (2024). Food Chain Partnership: When the Funding Stops: Sustaining private sector-non-profit collaborations. Case Study Solution presented at the International Food and Agribusiness Management Association (IFAMA), June 17, 2023. Christchurch, New Zealand.

• **Ac-Pangan, W.**, Hendricks, N. (2023). Improving existing methods of IFSA supply forecasting. Selected paper presented at the 2021 NCCC-134 Annual Conference, Applied Commodity Price Analysis, Forecasting and Market Rick Management. April 24, 2023. St. Louis, MO. Available at: https://legacy.farmdoc.illinois.edu/nccc134/conf_2023/pdf/Ac-Pangan_Hendricks_Zereyesus_Kee_NCCC-134_2023.pdf

• **Ac-Pangan, W.** Coffey, B. (2021) How do changes in market fundamentals affect the hedging in US live cattle markets? Selected paper presented at the 2021 NCCC-134 Annual Conference, Applied Commodity Price Analysis, Forecasting and Market Rick Management. April 19-20, 2021. (Online). https://ageconsearch.umn.edu/record/316401

POSTER PRESENTATIONS

• **Ac-Pangan, W.**, Hendricks, NP. (2025). In-Season US Crop Acreage Forecasting Using Machine Learning. Selected poster will be presented at Agricultural and Applied Economics Association Meeting (AAEA) July 29, 2025. Denver, Colorado.

• **Ac-Pangan, W.**, Hendricks, NP. (2025). Effects of Non-Tariff Trade Barriers in Grain Markets: The Case of Rice Export Ban Imposed by India. Selected poster will be presented at Agricultural and Applied Economics Association Meeting (AAEA) July 29, 2025. Denver, Colorado.

• **Ac-Pangan, W.**, Sharma, P., Kim, J. (2024). Identifying the Untapped Potential of the Small Local Food Manufacturing Business in the US. Poster presented at AI and the Future Symposium: Trust AI?. Oct 15, 2024. Manhatan, Kansas.

• **Ac-Pangan, W.**, Hendricks, NP. (2024) Using Machine Learning to Forecast Long-Term Grain Supply of Low and Middle-Income Countries. Poster presented at AI and the Future Symposium: Trust AI?. Oct 15, 2024. Manhatan, Kansas.

• Jo, Francis. Reyes, P. Hendricks, NP., **Ac-Pangan, W.** (2023) A Unified Algorithm For Penalized Likelihoods In Model Selection Of Spatial-Temporal Models For Lattice Data. Research and the State, Graduate Student Poster Session. Kansas State University. April 20, 2023. Manhattan, KS. Available at: https://www.k-state.edu/grad/student-success/research-forums/docs/research-state/2023-RS-Abstract-BookleT.pdf

• Hallman, L., Rossi, L., **Ac-Pangan, W.**, Adams, S. N. (2019). Optimal Nutrient Concentrations and Use of Root Growth Enhancers to Improve Citrus Root Health. Poster presented at American Society for Horticultural Science (ASHS 2019 Annual Conference), July 24, 2019. Las Vegas, Nevada. Available at: https://ashs.confex.com/ashs/2019/webprogramarchives/Paper30104.html

• Adams, S. N., **Ac-Pangan, W.**, Rossi, L. (2019). Effects of Soil Salinity on Citrus Rootstock 'US-942' Physiology and Anatomy, HortScience horts, 54(5), 787-792. Poster presented at American Society for

Horticultural Science (ASHS 2019 Annual Conference), July 24, 2019. Las Vegas, Nevada. Available at: https://ashs.confex.com/ashs/2019/webprogramarchives/Paper30091.html

EXTENSION ACTIVITY

Department of Agricultural Economics, Kansas State University
Assistant Extension Coordinator Aug – Dec 2024
76th Annual Kansas Income Tax Institute
Logistic Support for Online and In-Person Sessions (Supervised by Dr. Rich Llewelyn)

GRANT ACTIVITY

Title: AI-Driven Tool to Improve Decision-Making for Dairy Farmers Interested in Adopting Automation
Status/Amount: Proposal Submitted (Under Review) / $100,000
Source: USDA ARS AI Innovation Fund (FY25/26)
Investigators: Elizabeth French (PI), Luis Pena-Levano, **Walter Ac-Pangan**

TEACHING EXPERIENCE

Department of Agricultural and Applied Economics, Virginia Tech
Instructor Aug – Dec 2025
AAEC 4514 – Advanced Agribusiness Marketing
(45 Undergraduate students). Senior/Junior Level Course

Instructor Aug – Dec 2025
AAEC 3504 – Marketing Agricultural Products
(80 Undergraduate students). Junior/Sophomore Level Course

Department of Agricultural Economics, Kansas State University
Instructor Aug – Dec 2024
AGEC 115 – Decision Tools For Agricultural Economics and Agribusiness
(20 Undergraduate students). Freshman Level Course

Instructor Aug – Dec 2021
AGEC 632 – Agribusiness Logistics
(22 Undergraduate students). Senior Level Course

Instructor
R Programming Language Camp for first-year M.S. and Ph.D. students.
(10 Graduate students) Aug 2023
(14 Graduate students) Aug 2024

Graduate Teaching Assistant Aug – Dec 2019
AGEC 615 – Global Agricultural Development
(56 Undergraduate students). Senior level Course
AGEC 120 – Agricultural Economics and Agribusiness

(17 Undergraduate students). Freshman level Course

Department of Grain Science, Kansas State University
English - Spanish Translator and Interpreter Jun – Aug 2021
Educational Program USSSEC – International Grain Program at Kansas State University Practical and Strategic Education for the Next Generation of Agribusiness Leaders Online Course Taught in Mexico and Colombia

COMMETTEES AND SERVICE EXPERIENCE

Faculty Advisor, Undergraduate Curriculum 2025 - present
Department of Agricultural and Applied Economics, Virginia Tech

Faculty Advisor, Ag Econ Club and NAMA 2025 - present
Department of Agricultural and Applied Economics, Virginia Tech

INDUSTRY EXPERIENCE

Oro del Pacifico S.A. (Guatemala) 2018
Production Assistant Manager of Shrimp Farms

QualiTech Company, (Chaska, Minnesota) 2017
Lab Research Assistant in the AG Nutrition Department

LEADEARSHIP AND AWARDS

- KSU - Graduate Student Council Travel Award
  (Manhattan, Kansas. 2025)
- Social Chair at AAEA-Latin American Section
  (New Orleans, Louisiana. 2024 - 2025)
- AAEA / USDA AMS Local Food Economics Data Visualization Challenge Award (First Place)
  (New Orleans, Louisiana. 2024)
- IFAMA Case Study Competition Award (First place Graduate Level Category)
  (Christchurch, New Zealand. 2023)
- KSU - Graduate Student Council Travel Award
  (Manhattan, Kansas. 2023)
- Treasurer of the Graduate Student Agricultural Economics Association
  (Kansas State University, Manhattan, KS. 2023)
- First place in academic performance at Department of Agribusiness (*Summa Cum Laude*)
  (Zamorano University, Honduras. 2017)
- President of the Students Association at Department of Agribusiness
  (Zamorano University, Honduras. 2016 - 2017)
- Outstanding Academic Performance Award
  (Zamorano University, Honduras. 2015 - 2017)

PROFESSIONAL MEMBERSHIPS

Agricultural & Applied Economics Association (AAEA) 2021 - Present
Applied Commodity Price Analysis, Forecasting and Market Risk Management (NCCC-134) 2021 - Present
Australasian Agricultural and Resource Economics Society (AARES) 2024 - Present
Graduate Student Agricultural Economics Association at KSU (GSAE) 2019 - Present

International Food and Agribusiness Management Association (IFAMA) 2022 - Present
North American Colleges and Teachers of Agriculture (NACTA) 2024 - Present

SKILLS

• **Data Analytics:** Data wrangling, data visualization, multivariate statistical analysis, econometrics, machine learning, remote sensing and forecasting models, spatio-temporal and time series analysis, event studies, gravity models, risk analysis, and Monte Carlo simulations.

Software Proficiency:
• **R:** R Markdown (RMD), R Shiny (Dashboards).
• **Python:** pandas, scikit-learn, XGBoost, Keras, TensorFlow, statsmodels, NumPy, Matplotlib, Seaborn, Plotly.
• **STATA and SAS:** For data processing and modelling.
• **Document Preparation:** LaTeX, Overleaf, QUARTO (For reports and presentations).
• **Coding Environments:** Visual Studio Code, Google Colab, Jupyter, GitHub, RStudio, QUARTO.
• **@RISK:** Applied for risk analysis and simulation.
• **Microsoft Office Suite:** PowerBI, Excel, Word, PowerPoint, Access, Outlook.

Languages:

• English (Professional proficiency), Spanish (Native speaker).